iregi-rust benchmark

Batched one-loop tensor reduction in Rust versus IREGI and MadLoop

2026-10-23 · Timing environment not yet recorded

Reading guide. Every timing ratio is *iregi-rust time ÷ reference time*. Green means faster, white parity and red slower; timings are per phase-space point. Gray cells are unmeasured and amber cells are pending verification. Source inspection does not establish numerical agreement. The independent Fortran oracle passes its recorded initialization and identity checks. Rust/reference agreement and runtime performance remain unmeasured in this report.

Acceptance gates

Correctness	Agreement with original IREGI and isolated MadLoop graphs	**Pending**
Pure reduction	Equal or better performance on identical inputs	**Unmeasured**
Complete evaluation	Report preparation, numerator, reduction and masters separately	**Unmeasured**

1 Identical-input comparisons

Table 1: **Synthetic inputs versus original IREGI.** Each measured comparison must share routing, masses, scale, rank, numerator coefficients and reduction mode. The current cells contain no measurements.

Case	n / r	Reduction	Masters	Total
Bubble Explicit DOT numerator	2 / 0–3	**Unmeasured** vs IREGI	**Unmeasured** vs OneLOop	**Unmeasured** vs IREGI
Triangle Explicit DOT numerator	3 / 0–3	**Unmeasured** vs IREGI	**Unmeasured** vs OneLOop	**Unmeasured** vs IREGI
Box Expanded topology coverage	4 / planned	**Unmeasured** vs IREGI	**Unmeasured** vs OneLOop	**Unmeasured** vs IREGI



2 Physical diagram comparisons

Pending. Start with top-loop $gg \to H$ triangles and $gg \to HH$ boxes. FeynKit-generated physical numerators and graph-isolated MadLoop comparisons have not yet been measured. R2 and UV counterterms are excluded; bare UV/IR poles and R1 remain. A physical graph comparison must identify its model, routing, normalization, helicity and color configuration.

3 Validation coverage

Table 2: **Evidence ledger.** “Source audited” records code inspection; “verified” identifies executed checks described in that row.

Capability	Required evidence	Status
Independent reference oracle	Untouched/repaired Fortran twins 24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass.	**Verified**
Coefficient ordering and normalization	Native IREGI and MadLoop conventions Original Fortran interfaces and scalar wrappers inspected; numerical checks pending.	**Source audited**
Scalar-master backend	OneLoopMaster, all batch lanes Native evaluation integration and reference comparison pending.	**Pending**
Precision	binary64, DoubleFloat, arbitrary precision No numerical coverage claimed.	**Pending**
Automatic differentiation	Dual values and derivatives Derivative checks pending.	**Pending**
Vectorization	Batches propagated through masters Widths 1, 4, 8, 16, 32 and 64 planned.	**Pending**
Exceptional kinematics	Lane-local errors and precision rescue Threshold and small-Gram corpus pending.	**Pending**
Physical verification	Isolated MadLoop graph, no R2 / UV counterterms Bare poles and R1 must remain in the comparison.	**Pending**

4 Measurement protocol

Run release builds on one pinned CPU core with fixed thread counts and record hardware, operating system, toolchain, compile flags, dependency revisions, input hashes, seeds, warm-up, sample count and raw observations. Every result must identify whether caches are cold or warm. Report median and dispersion over repeated sufficiently long timed blocks; retain an output checksum.

Preparation comprises graph import, rules, tensor identities, coefficient extraction and evaluator construction. Time numerator evaluation, pure reduction, scalar masters and complete evaluation independently. Measure the complete total directly: adding separately instrumented components is only a diagnostic. Distinguish reference-library calls from subprocess and file I/O.

Use the same samples at batch widths 1, 4, 8, 16, 32 and 64 and report time per point. For precision comparisons, match the numerical precision and include rescue costs. Record failed lanes; do not omit unfavorable points silently. Absolute/scaled coefficient errors, high-precision adjudication and derivative checks accompany performance results. No speed gate passes without correctness. The stable binary64 target is a mixed absolute/relative tolerance of 10^{-10} on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput separately for each topology/rank/mass family. Record IBP, PaVe and hybrid mode independently. The eventual reference envelope is seven propagators and rank seven, including generalized powers and dimension shifts.

5 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `typst compile docs/benchmark.typ docs/benchmark.pdf`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst’s `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

6 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

6.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order (p^0,p^1,p^2,p^3). Propagators are $D_j = \left(q+p_j\right)^2 - m_j^2 + i0$, raised to positive integer powers. Generalized integrals use $d = 4 + \mathrm{idim} - 2\varepsilon$ and the native raw measure; dimension shifts introduce powers of π. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes μ^2 to QCDLoop and μ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2) \le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

6.2 Native poles and MadLoop conversion

The native result order is

$$(U,I,J,F) = (\varepsilon_{\mathrm{UV}}^{-1}, \varepsilon_{\mathrm{IR}}^{-1}, \varepsilon_{\mathrm{IR}}^{-2}, \mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite}, \varepsilon^{-1}, \varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank r to MadLoop order (finite, simple pole, double pole). Defining $a = \gamma_E + \ln\pi$, $b = \frac{a^2}{2} + \frac{\pi^2}{12}$ and $c_r = \frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}}, S_{\mathrm{ML}}, D_{\mathrm{ML}}) = c_r(F + a(U+I) + bJ, U + I + aJ, J).$$

For a OneLOop-normalized scalar master (L_0, L_1, L_2), the scalar wrappers use

$$J = i\pi^2 L_2, \quad U + I = i\pi^2(L_1 - aL_2),$$

$$F = i\pi^2\left(L_0 - aL_1 + \left(\frac{a^2}{2} - \frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U,I,J,F) = (i\pi^2, -i\pi^2, 0, 0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

6.3 Polynomial and tensor interfaces

A rank bound R has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of q^0, q^1, q^2 in descending order; the exponent of q^3 is the remaining degree. Thus the start is $\left(1, q^0, q^1, q^2, q^3, (q^0)^2, q^0q^1, q^0q^2, q^0q^3, (q^1)^2, \ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

7 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

7.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.

7.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep D symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

7.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

Alwall et al., appendix C.1: original IREGI context.
HEPKit TensorReducer API.
Goode et al.: symmetry-orbit tensor projectors.
Collins and Matsumoto: Weingarten orthogonality recurrence.
NinjaRS benchmark: report typography, table and color-style reference.

Local audit targets: `references/gammaloop_feynkit/crates/feynkit-tensor/src/` (`reduction.rs`, `reduction/external.rs`, `orbit_projector.rs`, `weingarten.rs`); `crates/feynkit-graph/src/integrals.rs`. Native conventions: `references/madgraph/vendor/IREGI/src/` (`IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90`, `funlib.f90`).